

Review

A promising natural anthraquinones mediated by photodynamic therapy for anti-cancer therapy

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ABSTRACT

Background: Experimental studies emphasize the therapeutic potential of plant-derived photosensitizers used in photodynamic therapy. Moreover, several *in vitro* and *in vivo* research present the promising roles of less-known anthraquinones that can selectively target cancer cells and eliminate them after light irradiation. This literature review summarizes the current knowledge of chosen plant-based-photosensitizers in PDT to show the results of emodin, aloe-emodin, parietin, rubiadin, hypericin, and soranjidiol in photodynamic therapy of cancer treatment and describe the comprehensive perspective of their role as natural photosensitizers.

Methods: Literature searches of chosen anthraquinones were conducted on PubMed.gov with keywords: “emodin”, “aloe-emodin”, “hypericin”, “parietin”, “rubiadin”, “soranjidiol” with “cancer” and “photodynamic therapy”.

Results: According to literature data, this review concentrated on all existing *in vitro* and *in vivo* studies of emodin, aloe-emodin, parietin, rubiadin, soranjidiol used as natural photosensitizers emphasizing their effectiveness and detailed mechanism of action in anticancer therapy. Moreover, comprehensive preclinical and clinical studies on hypericin reveal that the above-described substances may be included in the phototoxic treatment of different cancers.

Conclusions: Overall, this review presented less-known anthraquinones with their promising molecular mechanisms of action. It is expected that in the future they may be used as natural PSs in cancer treatment as well as hypericin.

Introduction

Cancer is considered one of the most common diseases and the primary cause of death. According to the latest statistics, in 2023, it is estimated that almost 1 960 000 new cancer cases and around 610 000 cancer deaths will occur in the United States (Siegel et al., 2023). Malignant neoplasms kill around half of their victims. Around one-third of cancer cases are avoidable by eliminating risk factors such as smoking, obesity, being overweight, or eating unhealthy foods, as well as applying preventative approaches (Kubrak et al., 2022; Nasir et al., 2020; Siegel et al., 2022; Soerjomataram and Bray, 2021). Cancer risk may also be decreased by early identification, greater investment in current cancer

control strategies such as herpes simplex virus vaccination, proper care employing innovative methods, combination therapy, and targeted therapies. Worth emphasizing is the fact that care of mental and physical patients with cancer is very important for patients health. The most essential thing to remember is that many tumors may be cured if detected early and appropriately treated (Kubrak et al., 2022; Siegel et al., 2022).

Photodynamic therapy (PDT) is a quite new and minimally invasive therapeutic technique for treating non-oncological lesions and various cancerous and pre-malignant diseases (Dobson et al., 2018; Kwiatkowski et al., 2018). PDT is used for the cure of cancer such as skin cancer (Champeau et al., 2019; Woźniak et al., n.d.), cervical cancer (Afanasyev

Abbreviations: PDT, photodynamic therapy; PS, photosensitizer; MAP, mitogen-activated protein; ROS, reactive oxygen species; APAF1, apoptotic protease activating factor 1; IAPs, apoptosis inhibitors; PI3K, phosphatidylinositol 3-kinase; AKT, protein kinase B; SMAC, second mitochondria-derived activator of caspase; DIABLO, Direct Inhibitor of Apoptosis-Binding protein with LOW pI.

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et al., 2022; Warowicka et al., 2019), breast cancer (Banerjee et al., 2020; Ostańska et al., 2021), and gastrointestinal tract cancers (Ding et al., 2021; Karshieva et al., 2022; Yano and Wang, 2020). Photodynamic therapy is also utilized effectively in dermatology to treat psoriasis (Wang et al., 2022) or cutaneous leishmaniasis (Dimmer et al., 2019), and dental diseases (Comeau et al., 2022; Shi et al., 2021). The origins of PDT can be dated to thousands of years ago, in ancient Egypt, India, and China, when herbal medicine was combined with sunshine. Nevertheless, until the 1900s, PDT was not well-known (Mansoori et al., n.d.). PDT is characterized by the local exposure of illumination to the lesion after the delivery of a photosensitizer substance that accumulates only in target cells. Subsequently, a cascade of photochemical and photobiological reactions occurs within cells, causing the death of cancer cells (Dobson et al., 2018; Kwiatkowski et al., 2018). Photodynamic therapy has higher selectivity against tumor cells than traditional methods of treatment due to the use of photosensitizers that are preferentially targeted in tumor areas and the specific light irradiation of these areas (Correia et al., 2021a).

In recent years, there has been an increasing number of studies evaluating the possible anticancer effects of natural chemicals and herbal components, especially those of plant origin, since they seem hopeful in medicaments and have fewer adverse effects than chemicals. Recent research has shown that herbal extracts, including substances that are targeted to eliminate tumor cells, can be employed in a variety of cancer therapies. Nature is a significant source of therapeutic plants, and many pharmaceutically active chemicals extracted from medicinal plants have yet to be evaluated for photoactive characteristics. Many researchers suggest that natural compounds with photosensitizing properties can be extracted from plants and utilized as replacements for conventional photosensitizers (PSs) used in PDT. With phototoxic and photoactive substances can be found in a variety of plant families. Nevertheless, there are significant challenges for medical therapy with phototoxic and photo-genotoxic substances isolated from plants and herbal materials, including their safety profile, the requirement for regulatory permission, and the demonstration of comparable efficacy to synthetic photosensitizers (Mansoori et al., n.d.; Muniyandi et al., 2020). However, research of natural products are fundamental for the development of novel drugs.

The primary goal of this review is to outline underlying mechanisms and druggability of potent and less known anthraquinones that can be used as photosensitizers in PDT. The authors collected the results of experimental *in vitro*, *in vivo* data and clinical studies discussing the possible application of anthraquinones as chosen plant-based-photosensitizers in PDT. To systematically estimate the effectiveness of emodin, aloe-emodin, parietin, rubiadin and soranjidiol in the context of photodynamic therapy for cancer treatment, we conducted a comparative assessment with hypericin, which has already been investigated in clinical studies

Photodynamic therapy - basics and principles

The photodynamic therapy is based on the interaction between a photosensitizer (PS) and a light with a specific wavelength. PS is administered intravenously or locally to the patient and accumulates in cancer cells. Then, after a predetermined amount of time, PS is photo-activated by exposure to a particular wavelength of light. In consequence, irradiation causes molecular oxygen to receive energy from light. It causes the generation of reactive oxygen species (ROS) such as singlet oxygen ($^1\text{O}_2$), superoxide radical ($\text{O}_2^{\bullet-}$), hydroxyl radical ($\text{HO}^{\bullet}$), and hydrogen peroxide (H_2O_2). These cytotoxic products initiate a series of biochemical reactions leading to tissue damage or destruction of the targeted tissue (Bacellar et al., 2015; Correia et al., 2021b; Donohoe et al., 2019a). PDT has several advantages over other standard approaches to cancer treatment. This method has a direct cytotoxic effect on cancer cells and is less toxic to healthy cells. Because of its dual selectivity, PDT may be utilized directly and precisely in malignant cell

treatment. The innate ability of certain photosensitizers to preferentially accumulate in tumor tissue and light irradiation solely in the target tissue are the two key variables that contribute to PDT selectivity. Topical administration facilitates the selective accumulation of PS in the tumor since PS is given directly and solely to the lesions before treatment. On the other hand, when a photosensitizer (PS) is administered intravenously, it is critical that the PS display stability and persist in the circulation for long enough to permit its accumulation inside the tumor site. (Calixto et al., 2016).

Moreover, PDT also leads to the destruction of the tumor blood vessels. The oxidative stress is anticipated to be greater around the tumor blood vessels and at the tumor surface, where the photon flow is larger and stronger, leading to inadvertent necrosis (Calixto et al., 2016; Donohoe et al., 2019b). Another advantage of PDT is that it can increase anti-tumor immunity. It improves the long-term control of the treatment of the disease and has a significant prevalence over conventional treatments like surgery or chemotherapy, which can even be immunosuppressive (Donohoe et al., 2019b). What is important, photodynamic therapy usually costs less than other cancer possible treatments (Dos Santos et al., 2019). Unfortunately, despite the many advantages, there are also a few disadvantages. The main side effect is severe pain that can occur when using intense irradiation (Gholam et al., 2010). Another limiting issue is increased blood pressure after irradiation and skin problems, such as redness, stinging, swelling, or itching (Keller et al., 2022).

Photodynamic therapy – mechanism

As previously stated, the photodynamic reaction begins with light absorption by the PS, which initiates a sequence of photochemical reactions that result in the formation of cytotoxic products. Following light absorption, the PS has an excited singlet state. This excited state is extremely unstable, and PS can respond in one of two ways (Fig. 1). Type I reaction involves an electron or a hydrogen transfer atom between the molecule of the excited dye and the neoplastic tissue in which the photochemical reaction takes place. As a result of these reactions, a radical anion or substrate radical is generated. These radicals react with molecular oxygen, producing ROS, such as $\text{O}_2^{\bullet-}$, $\text{HO}^{\bullet}$, and H_2O_2 , which causes oxidative damage that can lead to biological lesions. The type II mechanism depends on an initial energy transfer from the triplet excited state of the PS to dissolved molecular oxygen O_2 , which is in its ground triplet state. The triplet state has a longer lifetime, which allows time for direct transfer. This energy transfer process leads to the creation of singlet oxygen ($^1\text{O}_2$) and the basic state of the PS (Bacellar et al., 2015; Kubrak et al., 2022; Lee et al., 2020).

Photodynamic therapy – cell death

Maintaining the balance between the proper condition of the body and the appearance of certain abnormalities, such as oxidative stress, is very important in the body. If there is no such balance, the mechanisms of cell death are activated. Apoptosis, autophagy, and necrosis are cell-killing mechanisms. They differ in the way they work, but the result is the same - cell death (Fig. 2). In photodynamic therapy, we can observe three different mechanisms of cell death depending on the treatment conditions and the photosensitizer used. The use of a high concentration of photosensitizer and/or a high dose of radiation causes necrosis. Moderate damage caused by appropriate light concentration and dosage will result in controlled cell death (e.g., apoptosis). Finally, slight injuries cause autophagy. Most likely, adaptive autophagy, which, depending on the level of damage, might trigger other controlled cell death pathways (Donohoe et al., 2019b).

Apoptosis may be induced as a result of photodamage caused by photosensitizer and light. Photodamage to mitochondria causes membrane permeabilization, resulting in leakage of cytochrome c into the cytosol where it binds apoptotic protease activating factor 1 (APAF1) to

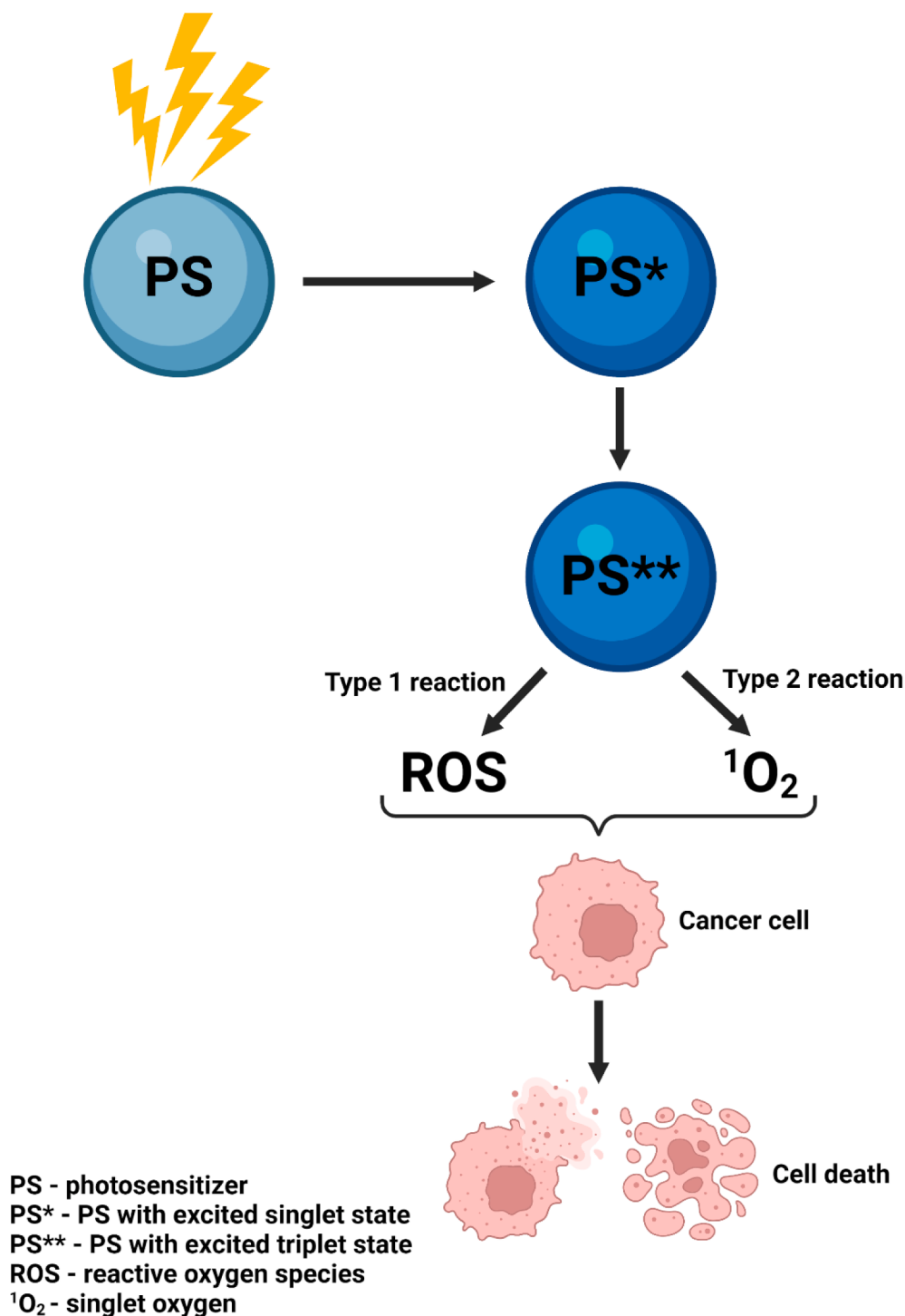


Fig. 1. Mechanism of photodynamic therapy.

form the apoptosome. This, in turn, activates the apoptotic pathway mediated by caspase-9. Apoptosis via mitochondria can be suppressed at varying degrees by anti-apoptotic proteins of the BCL family, including BCL2 and BCL-XL, and apoptosis inhibitors (IAPs), which are regulated by SMAC / DIABLO (a second activator of mitochondrial origin). Survival signals like growth factors and cytokines that stimulate the phosphatidylinositol 3-kinase (PI3K) pathway are other strategies to suppress apoptosis. PI3K stimulates AKT, which phosphorylates and inactivates a pro-apoptotic member of the BCL-2 family (Igney and Krammer, 2002; Kessel, 2019; van Straten et al., 2017).

Autophagy can act as a process responsible for cell death or a pro-

survival mechanism. The way of autophagy, depends on the PDT dose as with lower doses autophagy functions as a protective mechanism while with higher PDT doses, autophagic cell death can be initiated. Localization of the photosensitizer is also significant since it can either promote or inhibit autophagy. Lysosomal- and mitochondrial-targeted PS produce a pro-survival autophagic response. Atg7 is one of the numerous proteins that are involved in the autophagy process. The assembly of autophagosomes and the breakdown of molecules both include this protein (Donohoe et al., 2019b; Kessel and Oleinick, 2018; Mfouo-Tynga and Abrahamse, 2015; van Straten et al., 2017).

Necrosis is one type of cell death that frequently follows external

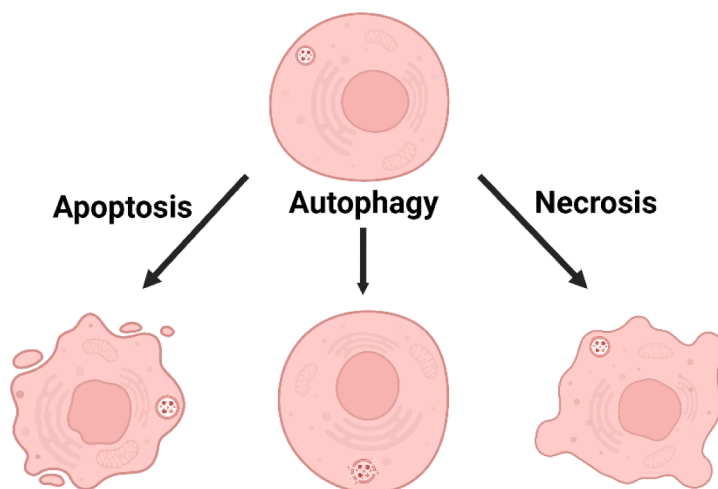


Fig. 2. Types of cell death after photodynamic therapy.

cellular damage. In photodynamic therapy, necrosis can happen after a high-dose medication or/and irradiation results in loss of membrane permeability and tissue necrosis. Following direct photodamage to the plasma membrane, necrosis might also take place (Kessel and Oleinick, 2018; van Straten et al., 2017). In necrosis, we can observe membrane permeability, movement of calcium ions across the endoplasmic reticulum, cytoplasmic swelling (oncosis), calcium-dependent calpain activation, lysosomal rupture, followed by degradation of cell components and induction of the inflammatory response (Mfouo-Tynga and Abrahamse, 2015).

Anthraquinones and their use in PDT

In recent years, photosensitive substances discovered in plants have been shown to exhibit a range of significant therapeutic and biological properties. One of notable group of such substances is anthraquinones. These compounds, including rhubarb and aloe, have been recognized and applied in traditional medicine for more than 4000 years. They are an important class of compounds with a wide range of applications. In addition to its usage as colorants, anthraquinone derivatives have long been utilized in medicine, including as laxatives, antimicrobials, and anti-inflammatory medicines. Current therapeutic indications include arthritis, multiple sclerosis, and cancer (Malik and Müller, 2016). This

study focuses on a specific group of substances, including emodin, aloe-emodin, hypericin, parietin, rubiadin, and soranjidiol, which possess photosensitizing properties and can act as an antiviral, antioxidant, and anticancer agents in photodynamic therapy. The biological, physical, and chemical characteristics of these compounds were investigated and comprehensively discussed here, shedding light on the biological mechanisms and potential applications of these compounds according to the results *in vitro*, *in vivo* studies, and clinical trials.

Methods

This review was prepared according to PRISMA (Preferred Reporting Item for Systematic Reviews and Meta-Analysis) guidelines. For literature search PubMed database was used. First, literature investigation was based on the key words of anthraquinones: “emodin”, “aloe-emodin”, “parietin”, “rubiadin”, and “soranjidiol” linked with “cancer”. Following results were linked with “photodynamic therapy”. Only studies assessing the effects of chosen compounds on cancer cell lines, mice were included in this article. Articles focusing exclusively on clinical trials involving the keyword “hypericin” were reviewed and the most promising results were described. The method of articles acquisition are shown in the flowchart (Fig. 3).

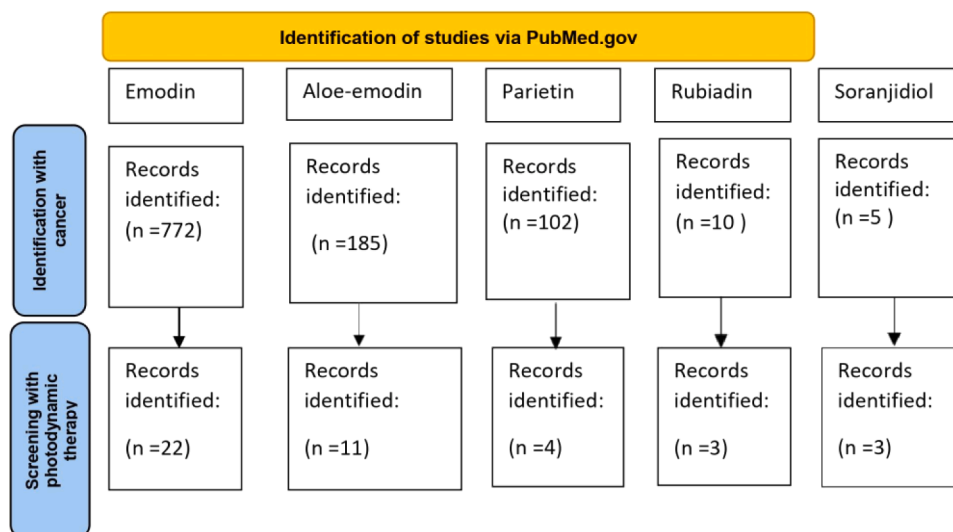


Fig. 3. PRISMA flowchart of study method. The procedure of literature searching with keywords.

Emodin

Emodin is a red-orange powder or crystal that belongs to the anthraquinone family. For over 2000 years, this phyto substance has been used in traditional Chinese medicine and may still be found in a range of herbal medicines. This substance was derived from a variety of Chinese medicinal plants, including *Radix rhizoma Rhei*, *Aloe vera*, *Polygonum multiflorum*, *Giant knotweed*, *Rheum palmatum*, and *Polygonum cuspidatum*. It is also present in the bark and roots of many other plants, molds, and lichens. Emodin possesses anticancer, anti-inflammatory, antioxidant, antibacterial, antiviral, anti-diabetes, immunosuppressive, and osteogenesis-stimulating properties (Dong et al., 2016; Semwal et al., 2021). According to Wang et al., emodin may transfer electrons to generate reactive oxygen species, which causes cell death after irradiation (Wang et al., 2021). Therefore, emodin may be used as a photosensitizing agent in photodynamic therapy (Galiardi-Campoy et al., 2021). Additionally, emodin induces necroptosis but not apoptosis and inhibits aerobic glycolysis, which both contribute to the death of cancer cells (Wang et al., 2021). Zhang et al. discovered an original approach by which emodin can suppress the growth of NSCLC cells by reducing the expression of sPLA2-IIa and NF- κ B activity. Moreover, emodin can inhibit the IKK/NF- κ B signaling pathway while promoting the AMPK pathway. It has been noted that emodin has a toxic effect on cell reproduction. As a result of the possible toxicity of emodin to healthy cells, its present applications are limited (Zhang et al., 2022). Galiardi-Campoy et al. (2021) investigated the effect of emodin combined with photodynamic therapy in cervical cancer cell lines, suggesting that it may be applied as a PS and shows an antitumor effect. The researchers studied two cervical cancer cells (SiHa and CaSki) and normal cells (HaCaT). The findings showed that emodin-PDT has anticancer activity through a variety of cellular and molecular mechanisms, including the induction of apoptosis and autophagy via ROS generation and differential expression of genes involved in apoptosis, autophagy, oxidative stress, the cell cycle, angiogenesis, migration, invasion, and metastasis formation. Nowak-Perlak et al. (2022) conducted research for the first time with emodin-PDT on skin cancer cell lines, including SCC-25 representing cutaneous squamous cell carcinoma, MUG-Mel2 representing a melanoma cell line, and normal human keratinocytes HaCaT representing control normal skin cells. To evaluate the effectiveness of emodin as a photosensitizer in PDT on different skin cell lines, they performed MTT assay measuring cytotoxicity of natural compounds, cellular uptake, apoptosis with flow cytometry, and a wound-healing assay. What they found, reveal that phyto substance affects skin cancer cells and causes growth-inhibiting or apoptotic death-inducing, and has a minimal cytotoxic effect on normal keratinocytes.

Aloe-emodin

Aloe-emodin is a novel anthraquinone substance derived from plants used in traditional Chinese medicine, including *Cassia occidentalis*, *Rheum palmatum* L., *Aloe vera*, and *Polygonum multiflorum* Thunb. Aloe-emodin is presently gaining popularity because of its unique anti-tumor effect on several tumor cells. Moreover, aloe-emodin has been shown to have a wide range of pharmacological effects, including anti-inflammatory, antibacterial, antiparasitic, neuroprotective, and hepatoprotective properties (Dong et al., 2020). Unfortunately, despite its many benefits, aloe-emodin's clinical applicability is limited due to low solubility, absorption, and unanticipated side effects for example nephrotoxicity (Kim et al., 2022; Shukla et al., 2017). Aloe-emodin exhibits light absorption capacity, therefore it may be applied as a photosensitizer agent (Liu et al., 2018). Aloe-emodin may induce apoptotic cell death via the mitochondria pathways. Also it can stop cell proliferation in G1 phase, while exhibiting high Bax expression and low Bcl-2 expression, resulting in apoptotic cell death. The anthraquinone can inhibit ubiquitin-mediated DUSP1 protein degradation and increase its expression, therefore inhibiting numerous cancer-associated

signaling pathways (Chen et al., 2018, 2023; Liu et al., 2018). Interestingly, our previous article showed that aloe-emodin has a greater cytotoxic effect than emodin on skin cancer cell lines. Evaluation of the retention time by chromatography showed that aloe-emodin appears to be less hydrophobic and is a stronger growth inhibitor than emodin after irradiation with blue light at a dose of 6 J/cm² (Nowak-Perlak et al., 2022). Lee et al. investigated the mechanisms of aloe-emodin-induced photocytotoxicity in lung cancer H460 cells. The scientists confirmed that aloe-emodin with irradiation promotes apoptosis of H460 cells based on the results of photoactivated aloe-emodin-induced cytoskeleton disruption. The actinin protein expression and mitogen-activated protein (MAP) kinase members is also related to photosensitized aloe-emodin-induced apoptosis. In the experiments of Lee et al., photoactivated aloe-emodin-induced cell death caused the mitochondrial permeability and a shift in apoptosis-related protein expression. Researchers also found that anoikis (type of apoptosis) caused by aloe-emodin with irradiation is mediated by the intrinsic and extrinsic death pathways in H460 cells in a caspase-dependent manner (Lee et al., 2010). Liu et al. investigated the effects of aloe-emodin on human oral squamous cell cancer *in vitro* and *in vivo*. The researchers observed changes in the morphology of oral mucosa cancer. When KB cells were examined under an optical microscope, aloe-emodin-mediated photodynamic therapy reduced cell migration. The cell proliferation was inhibited in G1 phase and the apoptosis increase was caused by reactive oxygen species generated from photoactivated aloe-emodin. The scientists noticed that phyto substance in photodynamic therapy presents high Bax expression and low Bcl-2 expression in cancer cells, thus causing cell apoptosis. In comparison, *in vivo* studies on mice have demonstrated that PDT is significantly more successful at reducing tumor development, and the persistence of AE in tumors following therapy may encourage death in KB cells (Liu et al., 2018). Lin et al. explored how emodin and photodynamic therapy affected a human gastric cancer cell line, proposing that it could be used as a PS and had antitumor effects. They conducted MTT assay to define optimal concentrations and illumination energy, and flow cytometry to evaluate the rate of cell death. The expression levels of caspase-9 and caspase-3 were determined by western blot analysis. The scientists showed that aloe-emodin-induced PDT demonstrated a significant inhibitive effect on the proliferation of human gastric cancer cells, and the upregulation of caspase-9 and caspase-3 protein levels may induce cell apoptosis (Lin et al., 2017). Tu et al. discovered that in osteosarcoma the main mechanism of apoptotic cell death, depends on the amount of aloe-emodin and the radiation exposure. They also demonstrated that the uptake of aloe-emodin was mainly by endoplasmic reticulum and mitochondria of MG63 cells aloe-. According to their research, the death of MG63 cells induced by aloe-emodin-PDT is triggered by ROS (Li et al., 2016). Tu et al. also investigated the same model and got similar results. They demonstrated that aloe-emodin concentration and light energy density influenced how efficiently therapy reduced the viability of human osteosarcoma MG-63 cells. The researchers confirmed that aloe-emodin-PDT induced the autophagy and apoptosis of human osteosarcoma cell line MG-63 through the activation of the ROS-JNK signaling pathway (Tu et al., 2016).

Parietin

Parietin or physcion is an anthraquinone pigment found as a secondary metabolite in lichens and plants, e.g. *Xanthoria parietina*, *Polygoni Multiflori Radix*, *Reynoutria japonica* Houtt, *Rumex japonicus* Houtt, *Rumex dentatus*, *Osmunda japonica* Thunb, also it is found in rhubarb. This anthraquinone is located as tiny extracellular crystals in lichens and plants. It is responsible for its strong orange-brownish or yellow-orange coloration, to protect plants from severe sun rays. Parietin has powerful pharmacological activity causing apoptosis, upsetting the cell cycle, slowing tumor spread, and decreasing chemoresistance. Furthermore, some studies reveal its excellent microbicidal effects against several

bacteria, viruses, and fungi. They also have anti-inflammatory, antioxidant, enzyme-inhibitory, lipid-regulating, and neuroprotective activities, according to several research (Avan and Akbulut, 2023; Comini et al., 2017; Xun et al., 2019). Phycion may induce apoptosis in cancer cells by activating AMPK signaling pathway. The activation of this pathway can trigger ER stress by increasing caspase-12 activation and p-PERK protein levels (Pan et al., 2016, 2018). The compound in the human intestine act as laxative compounds (Wianowska, 2014). Mugas et al. extracted parietin from the lichen *Teuloschistes nodulifer* (Nyl.) Hillman (Telochistaceae). They evaluated these plant substances as a potential photosensitizer on tumor cells employing UVA-Vis and blue light. The scientists showed that parietin and irradiation induce death of leukemic cells (K562), mammary tumors (LM2), ovary adenocarcinoma cells (IGROV-1), and keratinocytes (HaCaT). To indicate which mechanism of cell death occur after therapy, they conducted a cell cycle method that revealed a high percentage of sub-G1 cells (probably apoptotic) (Mugas et al., 2021a). Later investigations revealed that photodynamic therapy mediated by parietin decreased LM2 cell growth and migration *in vitro*. Following the topical application of parietin to mice with subcutaneous LM2 cell implants and the irradiation of blue light, researchers observed substantial tumor necrosis after treatment (Mugas et al., 2021b). Cogno et al. carried out research on monolayer and spheroids of human colorectal adenocarcinoma (SW480 cell line) and incubated the cells with different concentrations of the parietin and then irradiated them. They analyzed cell viability, nuclear morphology, and type of cell death. The researchers observed that parietin-based photodynamic therapy showed a significant cytotoxicity effect on the monoculture of colon cancer cells and demonstrated that parietin-PDT induced apoptosis in the cell line. In the experiments on spheroids, the application of parietin and irradiation presented similar results which were observed on monolayer (Cogno et al., 2020).

Hypericin

Hypericin, which belongs to the anthraquinone compounds is a secondary metabolite mostly found in the genus *Hypericum*, especially *Hypericum perforatum* L. or *St. John's wort*. Along with the advancement of research, hypericin exhibited great therapeutic possibilities for a promising therapeutic drug. Described a lot of the extraordinary advantages of this natural substance, for example, anti-cancer, antidepressant anti-virus, and anti-bacterial. Due to its hydrophobic structure, hypericin usually accumulates in the membranes of the endoplasmic reticulum (ER), lysosomes, golgi apparatus, and mitochondria. Moreover, it has negligible dark toxicity, high photosensitivity, and a maximum absorption at 590 nm, so it is a promising photosensitizer, as well as its photochemical properties in photodynamic therapy. Thanks to its photochemical properties it generates high singlet oxygen and other reactive oxygen species (de Andrade et al., 2021; Dong et al., 2021; Wu et al., 2023). Hypericin has also disadvantages. It has a low bioavailability because can create complexes with other flat, aromatic molecules. Additionally, hypericin is characterized by poor water solubility. It is caused by the formation of dimers and higher clumps of hypericin's in water. More and more, scientists create modifications of hypericin in order to avoid its disadvantages (Olek et al., 2023; Pietrzak et al., 2022, 2014). Based on the experiments results hypericin has anticancer activity because it could bind with DNA and inhibit thioredoxin reductase (Staničová et al., 2018; Zhang et al., 2016). Additionally, it induces apoptosis through caspase-dependent pathway (Piryaei et al., 2021). Lenkavská et al. suggested treating protein kinase C (PKC) regulators simultaneously with hypericin-mediated PDT. Numerous isoforms of the PKC family, which have pro- or anti-apoptotic properties, are known to exist. Cell survival and apoptotic response were assessed, together with the expression of protein kinase C (PKC and PKC). The treatment resulted in a marked drop in cell viability and an increase of apoptotic cell response (Lenkavská et al., 2020). The study by Popovic et al. aimed to explore the cellular and molecular influence of

HYP-PDT on normal primary human keratinocytes (Kc), melanocytes (Mc), and fibroblasts (Fb) in an *in vitro* tissue culture model which represented both the epidermal and dermal cellular compartments of human skin. All the human skin cell types responded differently to a variety of HYP-PDT dosages, according to a cell viability investigation. These outcomes were consistent with the morphological analyses, which showed that Fb and Mc underwent morphological modifications. However, there was no discernible difference in the intracellular ROS levels in Mc or Kc while Fb showed a considerable rise in ROS levels. Additionally, Fb and Mc cell lines showed early apoptotic activity (Popovic et al., 2015). Maslaňáková et al. concentrated on hypericin-based PDT effects on cell viability, oxidative stress, and the distribution of Bcl2 family members in human coronary artery endothelial (HCAEC) cells. The researchers results demonstrated that the treatment caused HCAEC cells to dysfunctional mitochondria, largely die by a necrotic form of death, increased oxidative stress, and distribution of Bcl2 family members (Maslaňáková et al., 2016). Kim et al. conducted a multicenter phase 3 randomized clinical trial. This experiment included 169 patients with early-stage mycosis fungoides–cutaneous T-cell lymphoma (MF/CTCL). The Researchers used synthetic hypericin ointment, 0.25%, activated with visible light on a patient skin as a safe and effective therapy. They found that topical hypericin PDT was more effective than placebo after 1 cycle of treatment for 6 weeks. The responses grew to 40% after 2 cycles, 49% after 3 cycles, and were seen in skin and plaque lesions. The majority of adverse events were minor cutaneous responses at the application site. The results of this randomized clinical trial show that synthetic hypericin PDT has a good safety profile and is effective in treating early-stage patch and plaque MF/CTCL (Kim et al., 2022).

Rubiadin

Rubiadin is largely derived from the *Rubiaceae* plants (*Rubia cordifolia* and *Heterophyllaea pustulata*). It is an important component of the Ayurvedic medical system in the treatment of many ailments. Rubiadin has aroused a lot of attention in recent years because of its good pharmacological properties such as anticancer, anti-osteoporotic, hepatoprotective, neuroprotective, anti-inflammatory, antidiabetic, antioxidant, antibacterial, antimalarial, antifungal, and antiviral activity. The *in silico* results show that Rubiadin has a high ligand potential for a wide spectrum of macromolecules, implying that it may interact with several different enzymes or proteins. Rubiadin can affect cell death of cancer cells by inhibiting the phosphorylation of NF- κ B p65 (Shi et al., 2020). The knowledge contribute the development and use of Rubiadin as a promising candidate for medication (Watroly et al., 2021). Investigation of rubiadin toxicity was estimated using the variable closest neighbor (vNN)-ADMET webserver. According to the program's unconstrained prediction model, rubiadin can cause drug-induced liver damage, mutations, and mitochondrial dysfunction. Other findings indicated that anthraquinone had no negative effects (Schyman et al., 2017). The study by Cogno et al. aimed to explore rubiadin in photosensitizing properties in photodynamic reactions. In cell viability method observed a decrease in colon cancer SW480 cell survival after being exposed to the combination of rubiadin followed by irradiation with different light doses. The scientists observed that rubiadin-mediated-PDT leads to necrosis. In experiments on spheroids, researchers observed the highest discoloration and minor viability percentage at the uppermost light dose with a standard concentration. A rubiadin-PDT effect on viability was increased proportionally to the rise in the light dose and not correlated to the concentration of the drug (Cogno et al., 2020). Rummie Vitar et al. conducted MTT assay on MCF-7c3 cell line with rubiadin-mediated PDT and showed smaller cell viability than therapy with rubiadin alone. The changes in cell morphology was observed after rubiadin-PDT. The cells showed appearance with swollen membranes and irregular cytoplasm. The cells treated Rubiadin-PDT had characteristics of apoptotic cells, but nuclear condensation and DNA fragmentation could indicate that the treatment

rapidly trigger the late apoptosis stage. Rubiadin-PDT significantly increased caspase-3 activation, early apoptotic signaling protein that resulted in caspase-3-mediated DNA internucleosomal fragmentation (Rumie Vittar et al., 2014). Comini et al. studied the potential function of rubiadin as a phototoxic agent against human breast cancer (MC-7c3 cell line). Researchers investigated cellular death by photoactivated rubiadin after singlet oxygen generation, where cell viability decrease is related to tumor cell uptake (Comini et al., 2011).

Soranjidiol

Soranjidiol comes in the form of a yellow amorphous powder. The presence of soranjidiol has been identified in the species *Morinda* and *Tarenna* of the *Rubiaceae* family, which is exclusively found in tropical climatic zones. Almost all of these species' plants have been used in traditional folk medicine to cure a variety of ailments, including skin infections, boils, arthritis, diabetes, hypertension, hepatodisease, and cancer (Chen et al., 2020; Chokchaisiri et al., 2021; Zeutsop et al., 2021). Soranjidiol displayed moderate to mild cytotoxicity against human cervical (HeLa), human colon (HT 29), and human breast (MCF-7) cell lines, according to Chokchaisiri et al. (2021). Additionally, its bark and wood are used to cure fever and as an antimalarial drug in northeastern Thai traditional medicine. This anthraquinone exhibits a wide range of biological actions, including antioxidant, antibacterial, antiparasitic, and cytotoxic properties (Chokchaisiri et al., 2021; Zeutsop et al., 2021). According to Ugwah-Oguejiofor et al. soranjidiol has inhibitory activity against B-Raf kinase in cancer cells. The protein is irresponsible for sending signals inside cells which are involved in directing cell growth (Ugwah-Oguejiofor et al., 2023). In order to study the side effects of soranjidiol, scientists conducted experiments on mice. After applying anthraquinone and light, inflammation and erythema were observed on

the mouse skin. Animals treated with soranjidiol and kept in the dark showed no abnormal behavior or signs at the skin level. Also the control group (dark and irradiated) did not show any symptoms. Interestingly, after subcutaneous administration, similar effects were observed as in skin administration. Only the anthraquinone and irradiation group had side effects that occurred after several hours (Núñez Montoya et al., 2008). Cogno et al. conducted a study on a monolayer and spheroids of human colorectal adenocarcinoma SW480. The researchers incubated the cells with different concentrations of soranjidiol and then irradiated them. The cell viability test showed a decrease in cell survival from approximately 20% to 50%. The preferred type of cell death induced by soranjidiol-mediated PDT was necrosis (Cogno et al., 2020). Comini et al. researched human breast cancer MC-7c3 cell line. The scientists used soranjidiol as a phototoxic agent in this study. The anthraquinone exhibited significant photocytotoxicity on cancer cells and the cell death was caused by singlet oxygen production after photodynamic therapy (Comini et al., 2011).

Discussion and conclusions

Cancer is one of the current world's most significant problems. Due to its benefits, photodynamic therapy is of interest not only as a way of treatment and diagnostics of oncogenic changes but also as a complementary approach used in combination with traditional treatments. In this review, anthraquinones, in particular emodin, aloe-emodin, hypericin, parietin, rubiadin, and soranjidiol, may be utilized as photosensitizers in PDT for the treatment of cancer. These natural substances derived from plants, may act as photosensitizers with less side effects than synthetic chemicals, and have the benefit of selective uptake by cancer cells. The phototoxic effect of phytochemicals that belong to this group of plants in different cancer treatments with the use of PDT is

Table 1
Anthraquinones in cancer prevention: natural occurrence, and application.

Plant Compound	Occurrence of plant substance	Type of cancer	Experimental model	Reference
Emodin	Plants (<i>Radix rhizoma Rhei</i> , <i>Aloe vera</i> , <i>Polygonum multiflorum</i> , <i>Giant knotweed</i> , <i>Rheum palmatum</i> , <i>Polygonum cuspidatum</i>), bark and roots of plants, molds, and lichens	cervical carcinoma cells	<i>In vitro</i> - SiHa, CaSki, and HaCaT cells line	(Galiardi-Campoy et al., 2021)
		skin cancer	<i>In vitro</i> - MUG-Mel2, SCC-25, and HaCat cells line	(Nowak-Perlak et al., 2022)
Aloe-emodin	<i>Cassia occidentalis</i> , <i>Rheum palmatum</i> L., <i>Aloe vera</i> , and <i>Polygonum multiflorum</i> Thunb	Lung cancer	<i>In vitro</i> - H460 cell line	(Lee et al., 2010)
		Oral squamous cell carcinoma	<i>In vitro</i> - KB cell line <i>In vivo</i> - Mice with tumor	(Y. Liu et al., 2018)
		Gastric cancer	<i>In vitro</i> - SGC-7901 cell line	(Lin et al., 2017)
		Osteosarcoma	<i>In vitro</i> - MG-63 cell line	(Li et al., 2016; Tu et al., 2016)
Parietin	<i>Xanthoria parietina</i> , <i>Polygoni Multiflori Radix</i> , <i>Reynoutria japonica</i> Houtt, <i>Rumex japonicus</i> Houtt, <i>Rumex dentatus</i> , <i>Osmunda japonica</i> Thunb, <i>rhubarb</i>	Human myelogenous leukemia, human ovarian cancer, mammary adenocarcinoma, and human keratinocyte	<i>In vitro</i> - K562 cell line, IGROV-1 cell line, LM2 cell line and HaCaT cell line	(Mugas et al., 2021a)
		Mammary adenocarcinoma	<i>In vitro</i> - LM2 cell line <i>In vivo</i> - mice with tumor	(Mugas et al., 2021b)
		Human colorectal adenocarcinoma	<i>In vitro</i> - SW480 cell line	(Cogno et al., 2020)
	<i>Hypericum hirsutum</i> , <i>Hypericum perforatum</i> L., <i>St. John's wort</i>	Human glioma and human breast ductal carcinoma	<i>In vitro</i> - U87 MG cell line and BT 474 cell line	(Lenkavskaja et al., 2020)
		Primary human melanocytes (Mc), primary human keratinocytes (Kc), and primary human fibroblasts (Fb)	<i>In vitro</i> - Mc cell line, Kc cell line, and Fb cell line	(Popovic et al., 2015)
Hypericin		Normal human coronary artery endothelial cells	<i>In vitro</i> - HCAEC cell line	(Maslanáková et al., 2016)
		Early-stage mycosis fungoides—cutaneous T-cell lymphoma (MF/CTCL)	Clinical trials	(E. J. Kim et al., 2022)
Rubiadin	<i>Rubia cordifolia</i> and <i>Heterophyllaea pustulata</i>	Human colorectal adenocarcinoma	<i>In vitro</i> - SW480 cell line	(Cogno et al., 2020)
		Human breast cancer	<i>In vitro</i> - MCF-7c3 cell line	(Rumie Vittar et al., 2014)
		Human breast cancer	<i>In vitro</i> - MCF-7c3 cell line	(Comini et al., 2011)
Soranjidiol	<i>Rubiaceae</i>	Human colorectal adenocarcinoma	<i>In vitro</i> - SW480 cell line	(Cogno et al., 2020)
		Human breast cancer	<i>In vitro</i> - MCF-7c3 cell line	(Comini et al., 2011)

currently the subject of many scientific studies (Table 1). Presented experimental studies carried out *in vitro* and *in vivo* indicate their significant potential to eliminate cancer cells selectively. Furthermore, the results presented in the articles indicate that all chosen natural photosensitizers can exert phototoxic effects in two ways. On the one hand, they destroy cancer cells through different kinds of cell death, i.e., apoptosis, necrosis, and autophagy. On the other hand, according to the advantages of PDT, they increase oxidative stress, improve immune system response, and destroy the vascular system in cancerous tissue. Although the use of natural substances in PDT shows promise, there are several restrictions and difficulties that must be overcome. The safety profile of phototoxic and photo-genotoxic compounds isolated from plants and herbal products is one of the main limits. Moreover, limitations of the review include specific cancer types where PDT with anthraquinones has been studied, such as skin cancer, cervical cancer, breast cancer, and gastrointestinal tract cancers. However, the research do not encompass other cancer types. Collected studies (mostly *in vitro*) do not have the results confirmed by clinical studies yet. While the review shows encouraging findings from preclinical investigations, putting these outcomes into regular clinical practice may create difficulties. For anthraquinone-based PDT to be used in practice, factors including accessibility, standardization, scalability must be taken into account. Definitely potential of the presented natural substances is significant, nevertheless more studies are needed to characterize those substances. Additionally, researchers should be focused on several objectives, such as anticancer, pharmacodynamic, and pharmacokinetic effects on animal models, data indicating the optimal and toxic doses, research regarding their potential side effects, and results evaluating the molecular pathways related to phototoxic action of natural compounds.

To conclude, there are currently no approved anthraquinones specifically used for photosensitizers. However, among the anthraquinone derivatives, hypericin stands out as a compound that is being investigated in clinical trials. It is worth highlighting that the use of anthraquinones in clinical settings is still under exploration and it is expected that in the near future many natural PSs will be used in cancer treatment as the experimental data support their potential role in photodynamic therapy.

CRediT authorship contribution statement

Martyna Nowak-Perlak: Conceptualization, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Piotr Ziolkowski:** Supervision. **Marta Woźniak:** Supervision, Funding acquisition, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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